

Functions



Function notation and evaluation

1 If $f(x) = 2x^2 - 3x + 1$, find $f(4)$.

2 If $g(x) = \frac{x+3}{x-1}$, find $g(-2)$.

Domain and range

3 Find the domain of $f(x) = \frac{1}{x-5}$.

4 Find the domain of $g(x) = \sqrt{x-3}$.

Composite functions

5 If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

6 If $f(x) = 3x - 2$ and $g(x) = x + 4$, find $(g \circ f)(x)$ as a simplified expression.

Inverse functions

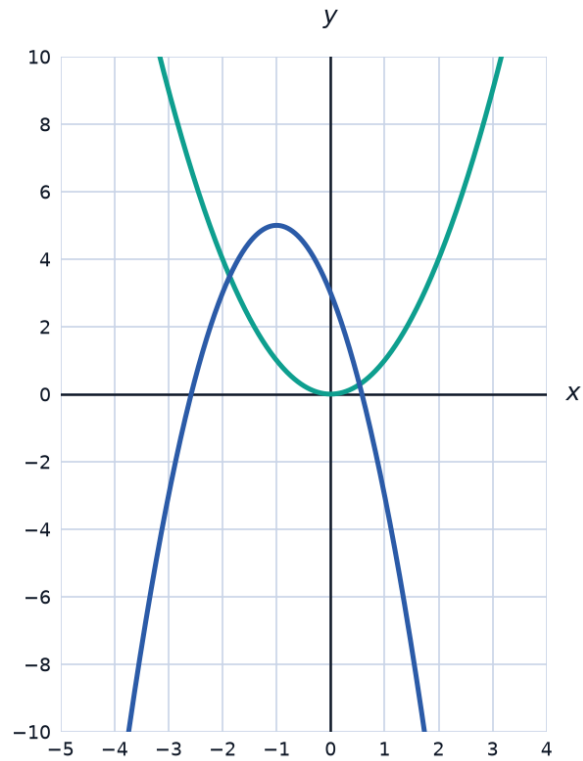
7 Find the inverse of $f(x) = 3x - 6$.

8 If $f(5) = 12$, find $f^{-1}(12)$.

Transformations of functions

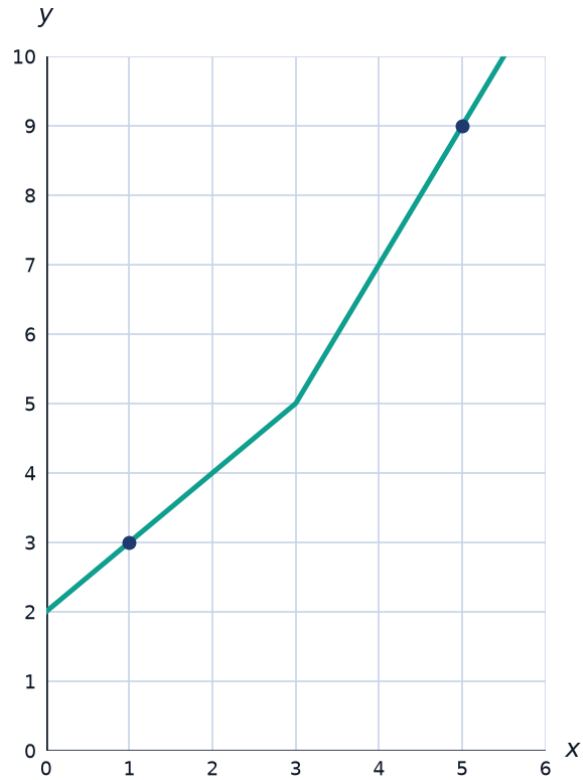
9 The graph of $y = f(x)$ is shifted 3 units right and 2 units up. Write the equation of the transformed function in terms of f .

- 10 Describe the transformation that takes $y = x^2$ to $y = -2(x + 1)^2 + 5$.



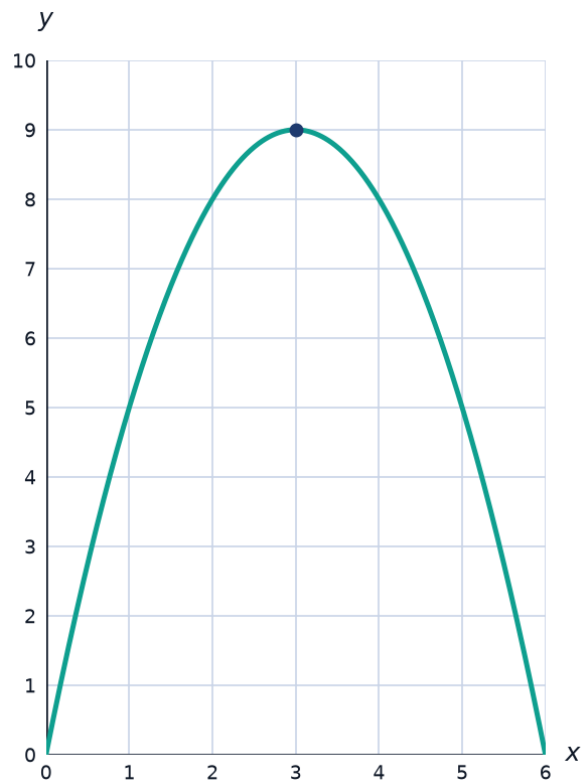
Piecewise functions

- 11 A function is defined as $f(x) = x + 2$ for $x < 3$ and $f(x) = 2x - 1$ for $x \geq 3$. Find $f(1)$ and $f(5)$.



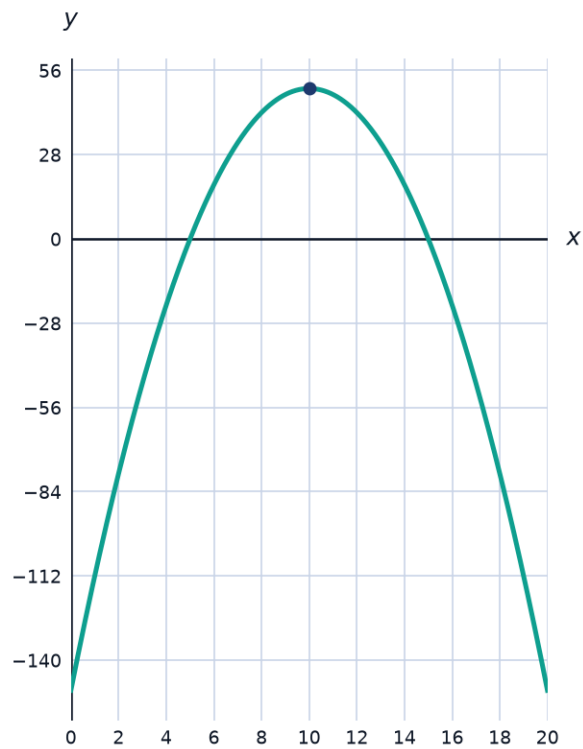
Interpreting function graphs and increasing/decreasing behavior

- 12 A function's graph rises from $x = 0$ to $x = 3$, reaches a peak at $x = 3$, then falls from $x = 3$ to $x = 6$. State the intervals where the function is increasing and decreasing, and identify the type of point at $x = 3$.



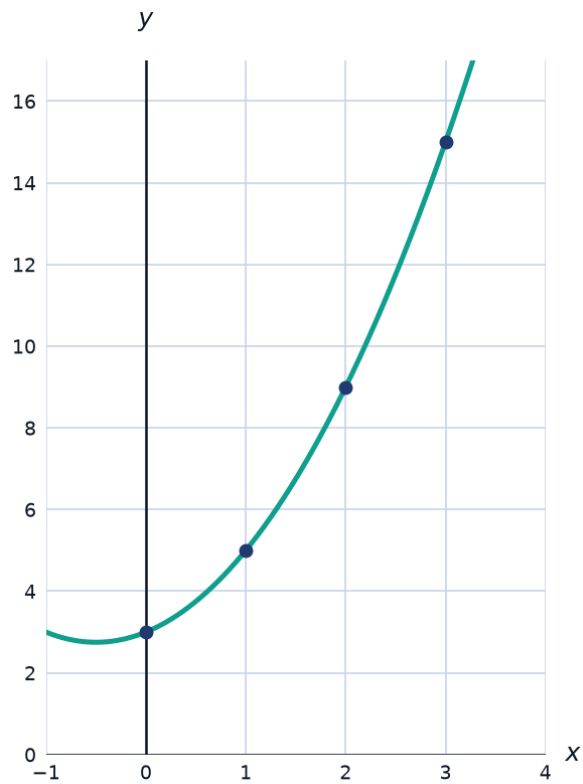
Modeling with function families

- 13 A company's profit is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the number of units sold (in hundreds). Determine whether this is a linear, quadratic, or exponential model, and find the number of units that maximizes profit.



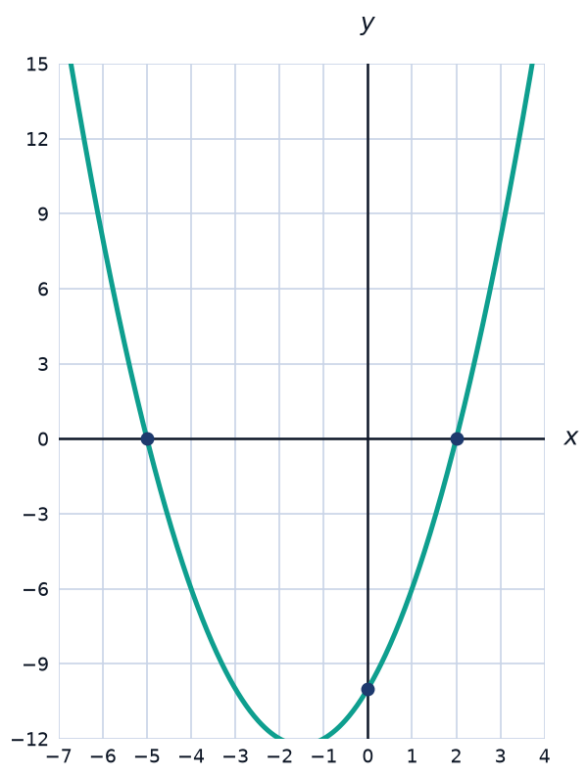
Finding function values from a graph or table

- 14 A table shows: $f(0) = 3$, $f(1) = 5$, $f(2) = 9$, $f(3) = 15$. Determine whether this could represent a linear or quadratic function, by examining the differences between consecutive values.



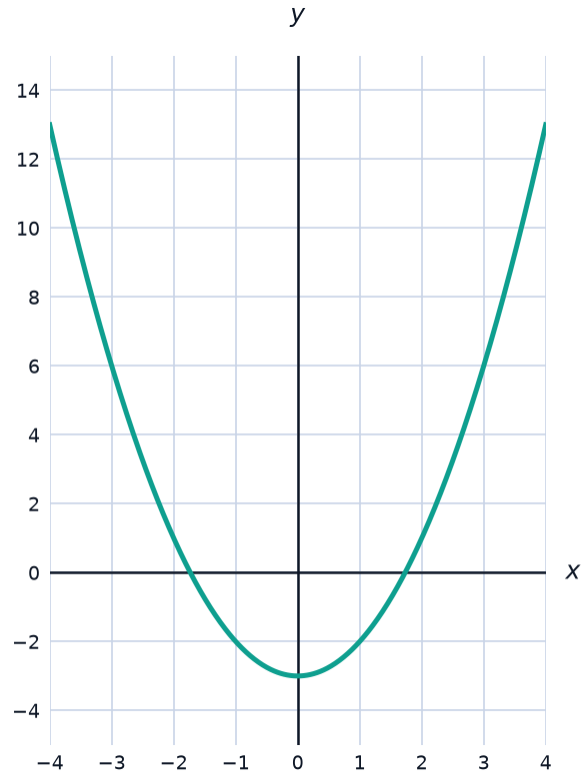
Zeros of a function and their graphical meaning

- 15 A function $f(x) = (x - 2)(x + 5)$ has zeros where it crosses the x -axis. Find the zeros, and state the y -intercept of the function.

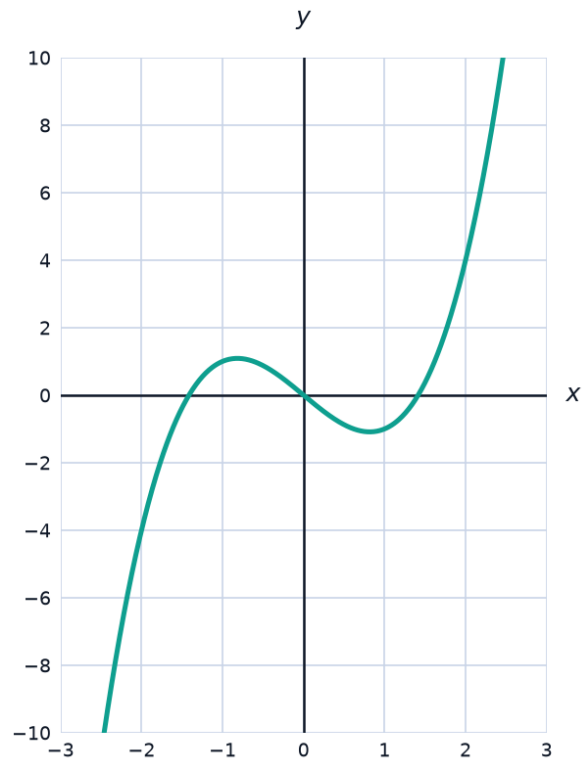


Even and odd functions

- 16 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, by evaluating $f(-x)$ and comparing to $f(x)$.



- 17 Determine whether $g(x) = x^3 - 2x$ is even, odd, or neither.



Modeling a real-world quantity with a function and evaluating it

- 18 The cost to rent a car is modeled by $C(d) = 35 + 0.20d$, where d is the number of miles driven. Find $C(150)$ and interpret the result in context.
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Solving for an input given an output

- 19 If $f(x) = 2x - 7$ and $f(a) = 15$, find the value of a .
- 20 If $g(x) = x^2 - 4$ and $g(b) = 21$, find all possible values of b .
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Multiple choice — easy

- 21 If $f(x) = 3x^2 - 5x + 4$, find $f(1)$.
- a. 2
 - b. 4
 - c. -6
 - d. 0
- 22 If $f(x) = 3x^2 - 2x + 5$, find $f(6)$.
- a. 101
 - b. 99
 - c. 103
 - d. 91

23 If $f(x) = -4x^2 + 4x + 7$, find $f(-6)$.

a. -161

b. -159

c. -175

d. -163

24 If $f(x) = 4x^2 - 5x - 9$, find $f(4)$.

a. 35

b. 53

c. 33

d. 37

25 If $f(x) = -x^2 + 5x - 1$, find $f(5)$.

a. 4

b. -3

c. -1

d. 1

26 If $f(x) = 2x^2 + 7x + 3$, find $f(4)$.

a. 57

b. 65

c. 63

d. 61

27 If $f(x) = -4x^2 + 3x + 1$, find $f(6)$.

a. -123

b. -125

c. -127

d. -120

28 If $f(x) = -x^2 - 5x + 1$, find $f(-1)$.

a. 10

b. 5

c. 3

d. 7

29 If $f(x) = 3x^2 + 4x + 3$, find $f(-3)$.

a. 18

b. 16

c. 12

d. 20

30 If $f(x) = 3x^2 + 9x + 7$, find $f(-1)$.

a. -13

b. -1

c. 3

d. 1

31 If $f(x) = -x^2 + 5x + 3$, find $f(5)$.

a. 1

b. 5

c. -3

d. 3

32 If $f(x) = -x^2 - 4x + 2$, find $f(-3)$.

a. 3

b. 1

c. 7

d. 5

33 If $f(x) = -2x^2 + 7x + 5$, find $f(-2)$.

a. -17

b. -15

c. -19

d. -27

34 If $f(x) = -x^2 - 5x - 3$, find $f(-5)$.

a. -5

b. 3

c. -3

d. -1

35 If $f(x) = 2x^2 + 2x + 9$, find $f(-6)$.

a. 67

b. 51

c. 69

d. 71

Multiple choice — medium

36 If $f(x) = 2x + 7$ and $g(x) = -2x - 7$, find $f(g(-3))$.

a. 5

b. -1

c. 3

d. 7

37 If $f(x) = -4x - 3$ and $g(x) = -3x + 5$, find $f(g(2))$.

a. 3

b. -2

c. -1

d. 1

38 If $f(x) = x + 9$ and $g(x) = x - 1$, find $f(g(-6))$.

a. 2

b. -7

c. 4

d. 0

39 If $f(x) = -4x + 6$ and $g(x) = -3x - 5$, find $f(g(2))$.

- a. 48
- b. 52
- c. -11
- d. 50

40 If $f(x) = 2x + 3$ and $g(x) = -2x + 1$, find $f(g(-4))$.

- a. 21
- b. 9
- c. 23
- d. 19

41 If $f(x) = -4x - 4$ and $g(x) = x - 7$, find $f(g(-1))$.

- a. 28
- b. -8
- c. 26
- d. 30

42 If $f(x) = -3x - 5$ and $g(x) = 4x + 7$, find $f(g(4))$.

- a. 23
- b. -76
- c. -72
- d. -74

- 43 If $f(x) = -4x - 1$ and $g(x) = x + 3$, find $f(g(6))$.
- a. -37
 - b. -39
 - c. -35
 - d. 9
- 44 If $f(x) = -x - 4$ and $g(x) = -2x + 8$, find $f(g(-5))$.
- a. -22
 - b. -24
 - c. 18
 - d. -20
- 45 If $f(x) = -3x + 1$ and $g(x) = -3x + 4$, find $f(g(-2))$.
- a. -27
 - b. -29
 - c. -31
 - d. 10
- 46 If $f(x) = -4x - 9$ and $g(x) = 4x + 1$, find $f(g(1))$.
- a. -31
 - b. 5
 - c. -27
 - d. -29

- 47 If $f(x) = 3x - 3$ and $g(x) = 4x + 7$, find $f(g(6))$.
- a. 31
 - b. 88
 - c. 90
 - d. 92
- 48 If $f(x) = -2x - 4$ and $g(x) = 3x + 2$, find $f(g(-2))$.
- a. 6
 - b. 4
 - c. -4
 - d. 2
- 49 If $f(x) = -3x - 6$ and $g(x) = -2x + 3$, find $f(g(-2))$.
- a. 7
 - b. -25
 - c. -29
 - d. -27
- 50 If $f(x) = 4x + 5$ and $g(x) = -x - 2$, find $f(g(-1))$.
- a. -1
 - b. 1
 - c. 5
 - d. 3

Multiple choice — hard

51 Find the inverse of $f(x) = -5x - 12$.

a. $\frac{x + 14}{-5}$

b. $\frac{x + 10}{-5}$

c. $\frac{x + 12}{-5}$

d. $\frac{x - 12}{-5}$

52 Find the inverse of $f(x) = -3x - 10$.

a. $\frac{x + 10}{-3}$

b. $\frac{x + 8}{-3}$

c. $\frac{x + 12}{-3}$

d. $\frac{x - 10}{-3}$

53 Find the inverse of $f(x) = -4x + 1$.

a. $\frac{x - 1}{-3}$

b. $\frac{x - 3}{-4}$

c. $\frac{x + 1}{-4}$

d. $\frac{x - 1}{-4}$

54 Find the inverse of $f(x) = -5x + 12$.

a. $\frac{x - 14}{-5}$

b. $\frac{x - 10}{-5}$

c. $\frac{x - 12}{-5}$

d. $\frac{x + 12}{-5}$

55 Find the inverse of $f(x) = 5x + 9$.

a. $\frac{x+9}{5}$

b. $\frac{x-7}{5}$

c. $\frac{x-11}{5}$

d. $\frac{x-9}{5}$

56 If $f(x) = -5x + 9$ and $f(t) = -36$, find the value of t .

a. 10

b. 8

c. 9

d. 11

57 If $f(x) = 5x - 4$ and $f(t) = -44$, find the value of t .

a. -8

b. -6

c. -9

d. -7

58 If $f(x) = -3x + 6$ and $f(t) = -18$, find the value of t .

a. 7

b. 10

c. 9

d. 8

- 59 If $f(x) = -2x + 6$ and $f(t) = 8$, find the value of t .
- a. 1
 - b. -2
 - c. -1
 - d. 0
- 60 If $f(x) = 2x$ and $f(t) = -14$, find the value of t .
- a. -6
 - b. -5
 - c. -7
 - d. -8
- 61 Which value of x is NOT in the domain of $f(x) = \frac{1}{x+8}$?
- a. -9
 - b. -8
 - c. -7
 - d. 8
- 62 What is the smallest value of x in the domain of $f(x) = \sqrt{x+4}$?
- a. -3
 - b. -5
 - c. 4
 - d. -4

- 63 Which value of x is NOT in the domain of $f(x) = \frac{1}{x-2}$?
- a. 2
 - b. -2
 - c. 1
 - d. 3
- 64 What is the smallest value of x in the domain of $f(x) = \sqrt{x-9}$?
- a. 8
 - b. -9
 - c. 9
 - d. 10
- 65 What is the smallest value of x in the domain of $f(x) = \sqrt{x-7}$?
- a. -7
 - b. 8
 - c. 6
 - d. 7